

Perovskite Photovoltaics in July 2026: Trap density and ion migration, self-assembled monolayer contacts

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Abstract

Across 648 indexed publications evaluated this month, with 181 examined in depth, the verified performance ceiling reaches a power conversion efficiency (PCE) of 33.1% in monolithic tandems and 27.31% in single-junction inverted devices. Tandem benchmarks rely on current matching across complementary bandgaps, whereas single-junction records depend on multidentate chelating agents and self-assembled monolayer (SAM) contacts that passivate undercoordinated lead cations and align interfacial dipoles. Yet only 22 closely read studies provide independently certified efficiency values, as uncertified reports frequently mask mobile ionic defects and non-radiative recombination under transient voltage sweeps. Scaling introduces severe resistive and geometric transport losses, establishing an area penalty that restricts certified efficiency to 27.31% at the largest reported aperture area of 766 cm². This scaling deficit arises because molecular passivators that eliminate trap states on microscale cells aggregate during meniscus drying, intensifying non-radiative recombination across laser-scribed module interconnects. Operational durability evidence remains narrow across the literature: only 4 investigations quantify lifetime metrics, with the longest duration reaching 9800 h, while only 6 studies adhere to standardized International Summit on Organic Photovoltaic Stability (ISOS) protocols. A pronounced reporting disparity persists between claimed and verified performance, with 50.4% of abstracts reporting efficiency but only 8.3% documenting independent certification. Mechanistic evaluation demonstrates that contact stabilization through self-assembled monolayers (SAMs) and multidentate coordination fails to establish resilience against thermo-mechanical fatigue, as thermal expansion mismatches induce cyclic shear stresses that trigger interfacial delamination. Depth-resolved degradation profiles confirm that breakdown initiates at buried selective contacts before penetrating bulk absorbers. Chemical treatments and interlayers also fail to achieve permanent suppression of halide phase segregation in wide-bandgap absorbers, where illumination and thermal cycling drive phase demixing without resolving the boundary between thermodynamic stability and kinetic demixing arrest. Finally, unstandardized aging without continuous tracking leaves the division between reversible ionic polarization and irreversible lattice degradation unquantified.

Keywords: perovskite photovoltaics; buried interface; defect passivation; wide-bandgap perovskite;

tandem solar cells; operational stability

1. Introduction

Perovskite photovoltaics has reached a juncture where incremental gains in laboratory power conversion efficiency (PCE) no longer mark material progress. The defining frontier has shifted to independent verification, scalability across module-relevant area, and operating lifetime under operational conditions. Peak metrics obtained from uncertified, sub-millimetre devices under transient voltage sweeps frequently obscure mobile ionic defects and non-radiative interfacial recombination that manifest only over extended timescales.

Within this landscape, 648 indexed works met the scope gate in July 2026 and 181 were read closely.

The mechanistic threads followed in this issue encompass passivation chemistry and ion migration under bias, self-assembled contacts and interfacial dipoles, precursor chemistry and phase purity, stability under load, heat and illumination, and wide-bandgap subcells and recombination layers. Together, these processes govern whether energetic band alignment and chemical defect suppression survive the coupled electrochemical gradients of field operation.

To evaluate these developments, this review distinguishes what the month’s literature verified from what it asserted. Filtering out artifacts of fast scan rates, uncalibrated spectral illumination, and negligible active areas isolates the specific chemical and structural interventions that provide authentic stability under working conditions.

2. The Certified Frontier in July 2026

The verified performance frontier reaches a power conversion efficiency (PCE) of 33.1% in two-terminal (2T) tandem architectures and 27.31% in single-junction inverted (p-i-n) devices, yet only 22 of 169 closely read papers reported an independently certified efficiency. In 2T tandems, certified values of 33.1% and 32.13%, along with intermediate benchmarks at 29.57%, 28.84%, and 27.49%, confirm that current matching across paired subcells dictates the verifiable ceiling [1,2,3,4,5]. Across p-i-n single junctions, certified PCE concentrates tightly at 27.31% and 27.3%, establishing an interfacial recombination barrier where non-radiative losses at transport layers suppress open-circuit voltage [6,7,8].

Area expansion reveals how uncertified laboratory reports outpace verified metrics. For a 625 cm² aperture, self-reported efficiency reached 27.8% despite an independently certified baseline of 27.3% [8]. Certified large-area devices quantify the resistive and geometric penalties imposed by macroscopic transport: 818 cm² achieved 26.53% [9], 766 cm² sustained 27.31% [7], 684.75 cm² reached 27.31% [6], and 655.2 cm² delivered 27.12% [10]. These metrics verify that scalable interconnects and spatial uniformity can suppress series resistance across large modules.

A champion efficiency obtained from an unshadowed, sub-0.1 cm² laboratory cell and an independently certified measurement across a module represent irreconcilable categories of physical evidence. Microscale test structures probe localized carrier generation under negligible series resistance, systematically masking pinhole shunting, film non-uniformity, and perimeter recombination. Conversely, certified module-scale metrics integrate transparent electrode sheet resistance, scribe dead zones, and macroscopic phase segregation across active layers. Equating isolated laboratory spikes with certified module performance mistakes local material potential for verified device technology.

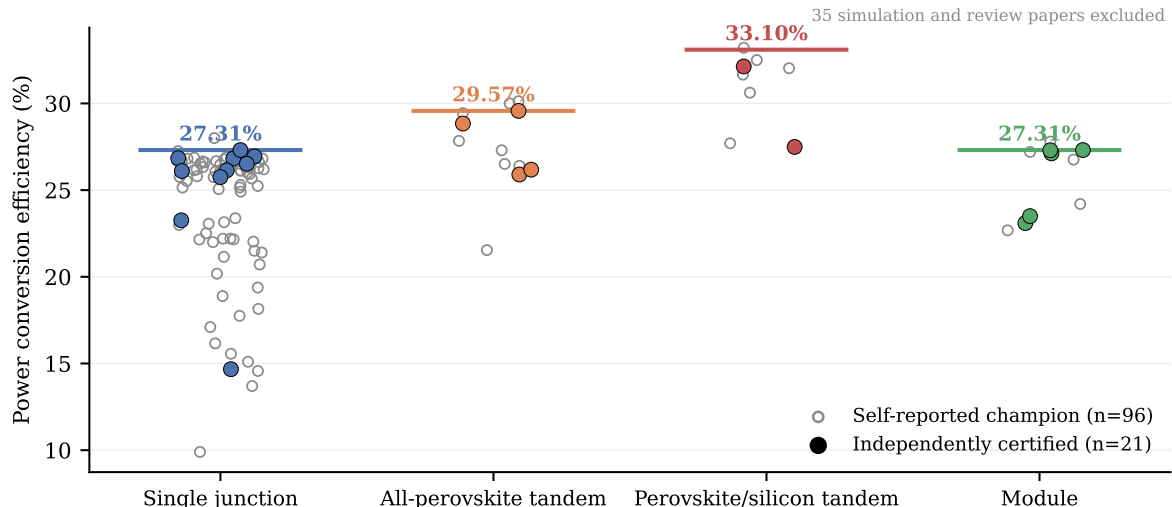


Figure 1. The certified frontier in July 2026. Filled markers are independently certified efficiencies, open markers are self-reported champion values, grouped by device family. Horizontal bars mark the highest certified value in each family.

3. Passivation Chemistry and Ion Migration Under Bias

The quantitative frontier for perovskite solar cells now reaches certified power conversion efficiency (PCE) values of 26.84% (champion 27.25%) in inverted devices and 23.5% on 65.05 cm² modules (champion 27.2%), though uncertified small-area claims reach 26.85% [11,12,13]. These benchmarks force a critical evaluation of what the reported physical evidence actually licenses: whether interfacial chemical treatments permanently suppress deep trap states and mobile ion channels under operating bias, or merely provide transient electrostatic screening that degrades during continuous operation.

At perovskite interfaces, undercoordinated Pb²⁺ cations and halide vacancies constitute primary non-radiative recombination centers. Monodentate Lewis bases donate electron pairs from sulfur, nitrogen, or oxygen to passivate surface Pb²⁺ defects, as demonstrated by carbazole-based self-assembled monolayers (SAMs) such as BTP at buried interfaces, thiophene sulfur coordination, and carboxyl/sulfur bifunctional TTA, which report uncertified champion PCE values of 26.85%, 21.15%, and 26.7% [13,14,15]. However, monodentate coordination offers limited thermal resistance against cooperative defect formation. To address this, amphoteric ionic liquids (GUHS and GUDHP) and FTAC simultaneously passivate Lewis acidic and basic sites while compensating iodine vacancies, enabling GUDHP to achieve 20.06% PCE with 85.74% fill factor (FF) at 1.84 eV bandgap (champion 26.13%) and allowing FTAC to retain 88.67% of initial efficiency after 1,0 h maximum-power-point tracking (maximum power point tracking (MPPT)) at 65 °C despite an uncertified 25.06% champion PCE [16,17]. Multivalent spatial anchoring further impedes defect formation: three-dimensional distribution of amino groups in tetrakis(4-aminophenyl)methane (4APM), nonlayered intermediate-site anchoring ligands, and m-bis(trifluoromethyl)aniline (mTFMA) bind PbI₂ or formamidinium iodide (FAI) precursors and lengthen carrier lifetimes from 529.7 to 2479.4 ns, retaining over 95% efficiency after 1000 h MPPT at 65 °C across 14 cm² (certified 26.26%) and over 90% PCE after 2000 h at 65 °C and 90% relative humidity (RH) (uncertified champion 26.54%, projected T80 of 9800 h) [18,19,20].

Under continuous bias, applied electric fields drive mobile iodide migration toward extraction layers, causing interfacial degradation and phase segregation. Chemical immobilization directly arrests this

migration: dual iodide-trapping agents (TFIBA and FPIFA), two-dimensional/three-dimensional (2D/3D) halide imidazolium heterojunctions, and 1-(4'-sulfophenyl)-3-carboxyl-5-pyrazolone sodium salt (SCP) layers suppress spiro-OMeTAD and electron-contact degradation, achieving an uncertified 26.81% champion PCE with T93 lifetime over 1140 h at 85 °C, 92% retention after 1200 h at 85 °C (champion 26.33%), and 90% retention after 1485 h MPPT (certified 26.84%, champion 27.25%) [11,21,22]. In wide-bandgap absorbers, field-driven halide separation is suppressed by guanidine phosphate (champion 21.54% at 1.78 eV, retaining 90.64% after 1400 h) and acetylthiocholine chloride (ATC) doping, which fills vacancies with quaternary ammonium and chloride ions to restrain demixing across 63.7 cm² modules (champion 22.68%) [23,24]. Conformal plasma surface engineering similarly mitigates defect accumulation at laser-scribed interconnects to secure a certified 23.5% module PCE on 65.05 cm² (champion 27.2%), with small-area devices retaining 98.1% of initial PCE after 2000 h MPPT at 85 °C [12].

Lattice strain fundamentally governs ion migration kinetics by lowering vacancy hopping barriers under tensile expansion. Relieving this strain through dual-site coordination with dithiopyr bridges adjacent Pb²⁺ sites, promoting preferential (100) crystal orientation and yielding a certified 26.14% PCE (champion 26.9%, and 21.28% on 30 × 30 cm² modules) [25]. Compressive strain engineering via the polymer MPUA (champion 24.91%), cross-linked 2-amino-3-hydroxybutanoic acid (AHBA) interfaces accelerating electron extraction to 3.5 ps (champion 26.32%), and photoresponsive C₆₀-azo trans-to-cis isomerization under ultraviolet (UV) light (92.7% retention after 488 h, champion 26.6%) counteract tensile distortion and impede defect formation [26,27,28]. Nevertheless, most strain and passivation studies report uncertified, small-area metrics without operando tracking of mobile ion accumulation under concurrent bias, heat, and illumination, leaving long-term field immunity unsubstantiated.

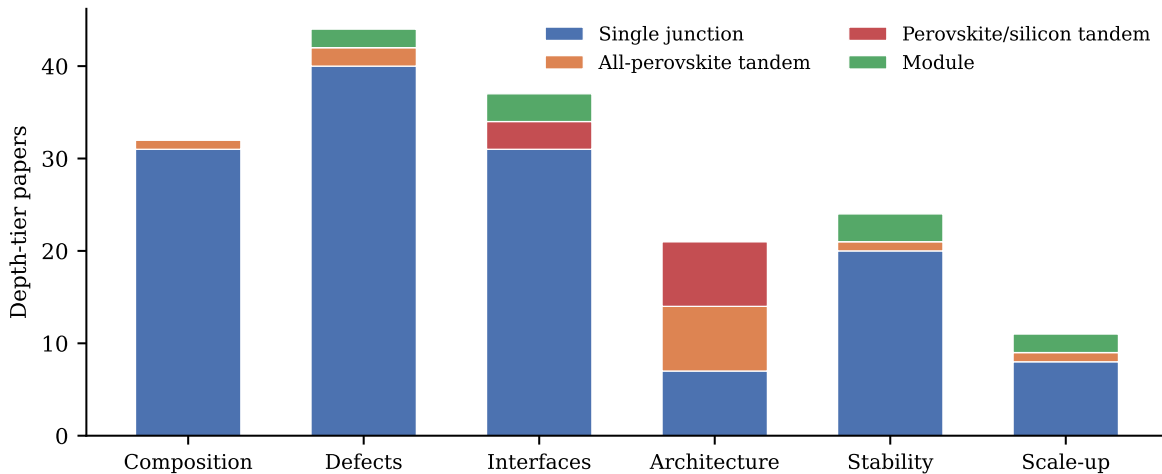


Figure 2. Where the month’s mechanistic effort was directed. Depth-tier papers by mechanism axis, stacked by device family.

4. Self-Assembled Contacts and Interfacial Dipoles

The quantitative frontier for contact-engineered perovskite photovoltaics is established by monolithic tandem configurations achieving certified power conversion efficiency (PCE) values of 33.1% (champion 33.3%) [1] and single-junction inverted architectures reaching certified 27.31% [6]. Translating

these efficiencies into durable devices hinges on resolving three core physical problems: weak oxide anchoring, self-assembled monolayer (SAM) aggregation, and interfacial dipole misalignment.

Substrate delamination and thermal desorption originate from fragile monodentate coordination at transparent conducting oxides. Mechanical toughness is markedly substrate-dependent: depositing self-assembled monolayers (SAMs) on a nickel oxide (NiO_x) underlayer increases interfacial fracture energy tenfold relative to indium tin oxide (ITO), while removing loosely bound molecules elevates fracture energy from 0.94 to 4.82 J m⁻² [29]. Multidentate chelating motifs directly counter this vulnerability by coordinating undercoordinated lead (Pb^{2+}) and reinforcing substrate anchoring: tetradentate iron(III) porphyrins yield certified 26.82% PCE (champion 27.06%) [30], bisphosphonates reconstruct the NiO_x /SAM buried interface to reach certified 27.31% PCE on small-area rigid cells [6], and zoledronic acid (ZDA) head groups yield 30.62% tandem PCE with wide-bandgap sub-cells (20.55% PCE) retaining over 90% performance after 200 h of maximum power point (MPP) tracking [6,30,31]. Lindqvist-type polyoxometalates similarly raise the anchoring ratio on ITO from 29.93% to 53.98% and cut the desorption rate from 30.7% to 5.6% during continuous MPP tracking (champion 25.15% PCE) [32]. Large-area module demonstrations remain scarce, though tetraphosphorylated phthalocyanines demonstrate a certified 23.1% PCE (stabilised 22.9%) on a 20.3 cm² module without degradation after 61 days of International Summit on Organic Photovoltaic Stability (ISOS)-O-1 outdoor aging [33].

Intermolecular pi-pi stacking among aromatic cores drives nanoscale SAM agglomeration, generating pinholes and phase inhomogeneities. Disrupting these interactions via steric hindrance or electrostatic confinement suppresses clustering: noncoplanar multidentate contacts (TTA) impede pi-pi stacking to stabilize wide-bandgap single-junction absorbers (approximately 1.67 eV, champion PCE of 23.7%) and tandems [1], while ionic liquid lubricants (BMIMBF₄) suppress aggregation via dual steric-electrostatic modulation, retaining 92.3% of initial efficiency after 800 h at 85 °C under air mass 1.5 global (AM 1.5G) illumination (uncertified champion 26.82% PCE) [34]. Concurrently, proton transfer complex-confined carbazole SAMs maintain thermal stability at 75 °C for 1368 h (champion 26.23% PCE) [35], and in situ thermally activated polymerization networks seal interfacial voids to achieve 26.87% champion PCE [34,35,36].

Unfavorable vacuum-level shifts and unpassivated defects dictate carrier extraction deficits at the absorber boundary. Halogenation and co-assembly modulate work functions: substituting methoxy groups with fluorine in F-2PACz strengthens the interfacial dipole and passivates halide vacancies, retaining 25.06% PCE on 1 cm² devices with 3.11% relative efficiency loss (champion 25.75%) [37], whereas co-assembling crown ethers at electron transport layer interfaces increases surface coverage from 0.57 to 0.79 and strengthens interfacial dipole moments from 2 to 7 Debye (champion 26.33% PCE) [38]. Polycyclic perylenecarbazole cores (6PCzBr) tailor electrostatic potentials for 26.20% PCE [39], phosphine additives (TFMP) coordinate Pb^{2+} and stabilize formamidinium (FA⁺) in Me-4PACz and 4PABCz ternary contacts (champion 26.78% PCE) [40], and ammonium bromide interlayers (PFN-Br) tune band alignment for 23.15% PCE [41]. However, because most dipolar gains are reported on uncertified sub-0.1 cm² cells without standardized bias protocols, whether these electrostatic barriers resist operational ion drift remains unverified.

5. Precursor Chemistry and Phase Purity

The quantitative frontier for perovskite photovoltaics is defined by single-junction power conversion efficiency (PCE) reaching certified values of 26.52% (26.88% champion) [42] and uncertified reports

exceeding 26% [43,44]. Whether these metrics reflect phase-pure lattices or metastable polymorphic mixtures remains governed by precursor kinetics, phase segregation, and spatial scalability. Record efficiencies frequently rely on sub-0.1 cm² apertures, such as 26.32% on 0.072 cm² [43], whereas scaling to a 900 cm² module lowers efficiency to 20.25% (17.75% certified) despite retaining over 91% initial efficiency after 1000 hours at 85°C and 85% relative humidity (RH) [42].

Controlling precursor coordination prevents solvent degradation and defective phase precipitation [45,46,47]. In formamidineum (FA) lead triiodide (FAPbI₃) growth, parasitic esterification between gamma-butyrolactone (GBL) and 2-methoxyethanol (2ME) destabilizes heated solutions; maintaining lead iodide (PbI₂) deficiency delays redissolution to suppress delta-phase impurities, yielding 22.99% PCE [45]. Additives also eliminate antisolvent quenching: sulfolane forms an intermediate phase yielding 22.2% PCE in methylammonium (MA) lead triiodide (MAPbI₃) [46], while 3-furoic acid (3-FA) coordinates tin iodide (SnI₂) and hydrogen-bonds with formamidineum iodide (FAI) to decouple nucleation from crystal growth, reaching 16.16% PCE [47]. Stoichiometric imbalance leaves defective binary halides: PbX₂-terminated surfaces display higher defect densities and lower stability than AX terminations [48]. Passivating residual PbI₂ with triphenyl phosphoramidate (TPPA) yields 21.4% PCE and retains 90% initial efficiency after 750 hours of maximum power point tracking (MPPT) versus 270 hours for controls [49]. Similarly, molten 4-(4,6-dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium chloride (DMTMM) dissolves two-dimensional (2D) phases to eliminate interfacial PbI₂, raising PCE from 23.43% to 25.73% [50], while optimizing PbI₂ evaporation distance achieves 14.57% PCE over 64 cm² [51].

Beyond nucleation, compositional heterogeneity drives phase demixing [52,53]. Concurrent four-dimensional scanning transmission electron microscopy (4D-STEM) and energy-dispersive X-ray spectroscopy (EDX) reveals that photoinactive hexagonal polytypes nucleate from nanoscale cesium (Cs) deficiencies [52]. In mixed-halide absorbers, strontium doping prevents I₂Br- intermediate formation, suppressing segregation to reach 20.18% PCE at 1.77 eV and 19.02% at 1.84 eV [54]. Rubidium A-site incorporation induces lattice distortion to suppress halide demixing, providing an open-circuit voltage (Voc) of 1.46 V in wide-bandgap sub-cells and 3.31 V in triple-junction tandems [55]. Conversely, trivalent bismuth substitution in cesium lead halides (CsPbX₃) triggers phase segregation due to lower thermodynamic stability [53]. Dielectric-response screening of 5043 compositions confirms that phase stability requires narrow windows of 2–8% Cs, 78–89% FA, 3–18% MA, and 93–99% iodide (I) [56].

Dimensional heterostructuring stabilizes interfaces, but phase purity remains constrained by transport penalties and area scaling [43,44,57]. In situ hydrogen-bonded hydroxyphenethylammonium (HPA) spacer networks retain 95.5% of initial efficiency after 3000 hours at 30°C in N₂, 92.5% in 45% ± 5% RH air, and 90% after 800 hours at 85°C in N₂, but on 0.072 cm² cells at 26.32% PCE [43]. Colloidal suspension epitaxy yields deterministic 26.0% multilayers without disturbing underlying layers [44]. In flexible tin cells, interfacial hydrogen bonding yields 15.56% PCE (14.67% certified) on 0.049 cm², but drops to 11.06% at 1 cm² despite achieving 22.21% under 1000 lux indoor illumination [57]. While precursor coordination and dimensional stabilization preserve microscopic phase purity, sub-0.1 cm² testbeds obscure severe interfacial penalties encountered during physical scaling.

6. Stability Under Load, Heat and Illumination

The quantitative frontier for operational durability is defined by time to 80% initial performance (T80) values of 9800 h [19] and 6785 h [58], falling steeply to 720 h [59] and 200 h [60]. However, this

frontier rests on narrow evidence: only 4 of 169 closely read studies report a T80 metric, and only 6 identify standardized International Summit on Organic Photovoltaic Stability (ISOS) protocols, such as ISOS-O-1 [33], ISOS-L-1 [7,61], ISOS-L-3 [62], ISOS-L-11 [9], or continuous maximum power point tracking (MPPT) [32]. Most stability claims instead rely on unstandardized, short-duration testing.

Depth-resolved profiling reveals that photo-induced degradation originates at both top and buried interfaces before penetrating the bulk [63]. To arrest interfacial chemical breakdown, molecularly tailored self-assembled monolayers (SAMs) stabilize buried contacts, yielding a certified power conversion efficiency (PCE) of 27.31% in single cells and 21.54% on a 766 cm² module under ISOS-L-1 [7], alongside asymmetric SAMs achieving 20.59% on 655.2 cm² modules without certified durability [64]. At the upper contact, multidentate fullerenes passivate defects and immobilize surface cations under ISOS-L-3, retaining 98.7% of initial PCE after 2100 h of continuous illumination and 95.8% after 1800 h at 85 °C, though on uncertified devices [62]. Interfacial defect passivation via aromatic carbonyls similarly claims stability but remains confined to an uncertified, sub-0.1 cm² cell of 0.09 cm² [65]. In-situ two-dimensional perovskite reconstruction further blocks interfacial degradation [66,67], while volatile dopants eliminate mobile lithium-ion residues in hole-transporting layers [68].

Under continuous bias and illumination, trapped-charge accumulation accelerates halide phase segregation, degrading wide-bandgap absorbers under standard MPPT, whereas maximum stability point tracking (MSPT) retains approximately 90% of initial power after 700 h [69]. Ion-migration barriers from diammonium additives suppress segregation to achieve a certified 26.18% PCE on a 1.17 cm² triple-junction aperture [70], aligning with rubidium doping to homogenize vertical cation distributions under ISOS-L-1 [61]. These degradation pathways are intensified by dynamic thermo-mechanical stress fluctuations [71], where inelastic tensile stress pre-relaxation reduces moisture ingress by up to an order of magnitude [72]. Without widespread adoption of standardized ISOS protocols, reported operational lifetimes remain poorly substantiated.

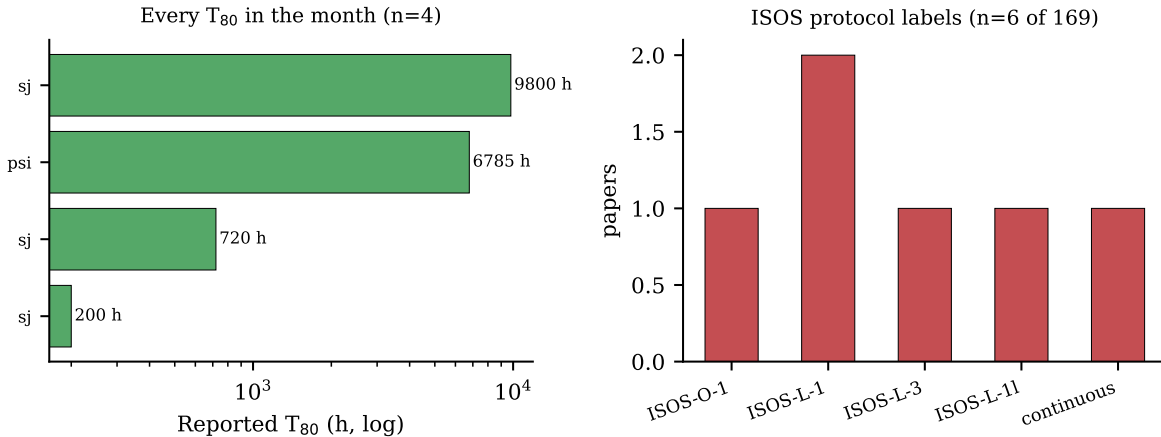


Figure 3. The month’s operational-stability evidence in full. Left, every reported time to 80% of initial performance. Right, the stability protocol labels used.

7. Wide-Bandgap Subcells and Recombination Layers

Monolithic perovskite-silicon tandems have pushed power conversion efficiency (PCE) to 33.21% (certified 32.13%) and 32.5% through interface and crystallization engineering [2,73], but reported performance exceeds what operating stability currently licenses. Two physical impediments dominate:

interfacial recombination at the hole-selective contact and halide phase segregation in wide-bandgap (WBG) absorbers.

At the buried contact, self-assembled monolayers (SAMs) suffer from solvent-induced degradation and inhomogeneous nucleation. Inserting a 3-TMA molecular interlayer between the self-assembled monolayer (SAM) and the absorber regulates crystallization via hydrogen bonding, reaching 23.17% PCE in 1.67 eV and 19.61% in 1.84 eV single-junction devices [2], while combining graphene oxide with SAMs forms a contact that retains 90% of initial efficiency after 770 h at 25 °C in a 27.3% tandem [74]. Coordinating formamidinium cations with gamma-valerolactone (GVL) cosolvents similarly retards nucleation to enforce oriented micrometer-scale grain growth across the buried interface [73,75].

Within the bulk, mixed-halide absorbers undergo light- and heat-induced phase segregation. Under in-operando thermal cycling from -100 to +100 °C, WBG subcells exhibit severe phase segregation above 50 °C with only partial recovery during dark cooling [76]. Halogen homogenization via TPB-Br surface passivation [77] and conductive polymer grain-boundary bridges [78] suppress halide demixing, yielding 23.63% (1.68 eV subcell) and 20.85% single junctions alongside uncertified tandems of 32.03% on 1.05 cm² and 25.92% [77,78]. Strengthening the metal-halide octahedral framework via Sr²⁺ alloying weakens electron-phonon coupling, achieving 31.66% PCE with an operational lifetime to 80% of initial efficiency (T80) of 6785 h on fully-textured silicon [58].

Upscaling these stacks exposes severe area-dependent transport penalties. Quantitative hyperspectral luminescence imaging on 64 cm² modules reveals that resistive transport and interconnection losses equal intrinsic material losses [79]. Scalable printing using low-toxicity GVL with dimethyl sulfoxide (DMSO) exceeds 17% in WBG cells and 28% in tandems [75], while ionic liquid stabilization demonstrates certified 27.49% tandems on 12.56 cm² with 23.58% certified top-cell efficiency [5].

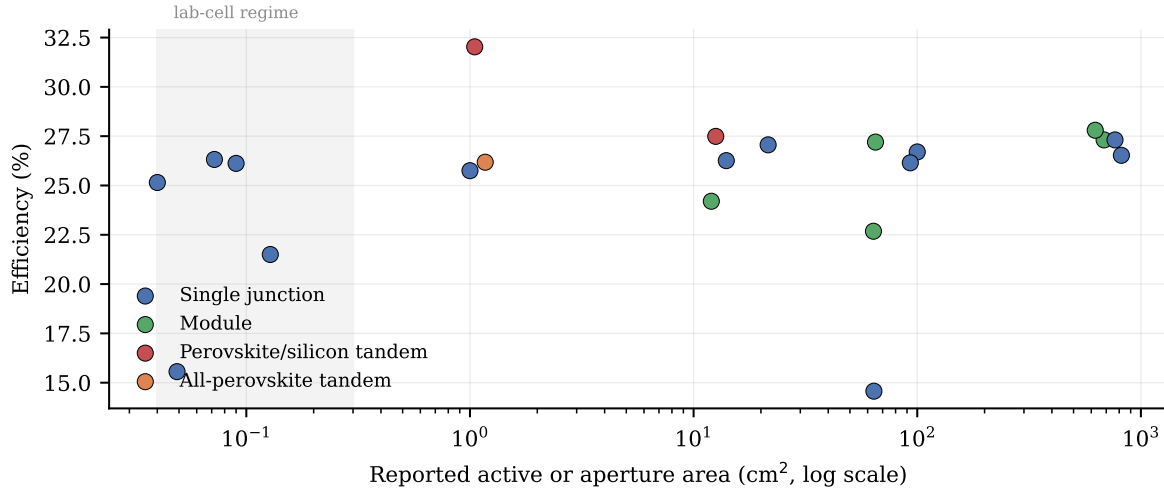


Figure 4. Efficiency against reported active or aperture area. The shaded band marks the sub-0.3 cm² laboratory-cell regime in which most champion values are measured.

8. Research Gaps and Outlook

The verified performance ceiling reaches a certified power conversion efficiency (PCE) of 33.1% in two-terminal (2T) tandems [1] and 27.31% in single junctions [6,7], yet the certified reporting share was 5.9% in 2026-06 and 8.3% in 2026-07, with only 22 of 169 closely read papers reporting any certified

value. These peak metrics do not license claims of operating durability, as uncertified microscale sweeps mask interfacial degradation, phase instability, and scaling penalties.

First, evidence for contact stabilization fails to establish whether interfacial dipoles and multidentate coordination withstand thermo-mechanical fatigue. While modifying transparent oxides with self-assembled monolayer (SAM) contacts and multidentate chelators elevates initial open-circuit voltage (Voc), thermal expansion mismatches induce cyclic shear stresses that trigger delamination [1,6,35]. Resolving whether coordination bonds survive operating fields requires operando double-cantilever beam fracture testing paired with bias-dependent Kelvin probe force microscopy during thermal cycling.

Second, reported chemical treatments do not demonstrate permanent suppression of halide segregation in wide-bandgap (WBG) absorbers. Although molecular interlayers and B-site alloying achieve high initial efficiency in tandems [2,58], in situ thermal cycling reveals phase segregation above 50 °C with only partial recovery upon cooling [76]. Differentiating true thermodynamic phase stability from kinetic demixing arrest demands operando nanoscale photoluminescence spectroscopy synchronized with diffraction profiling under continuous electrical bias and elevated temperature.

Third, unstandardized testing leaves the boundary between reversible ionic polarization and irreversible decomposition unquantified. Only 4 papers reported a time to 80% of initial performance (T80) lifetime, the longest 9800 h [19], and only 6 papers named an International Summit on Organic Photovoltaic Stability (ISOS) protocol [62,69]. Establishing ion accumulation rates under operating conditions requires operando intensity-modulated photovoltage spectroscopy coupled with secondary-ion mass spectrometry during maximum power point tracking (MPPT).

Fourth, microscale defect passivation fails to establish scalability across large apertures. Only 27 of 648 works addressed scale-up, and the largest reported aperture reached 818 cm² [9]. Passivators that eliminate traps on sub-0.1 cm² apertures aggregate during meniscus drying, intensifying non-radiative recombination across laser-scribed modules [42,79]. Spatially resolved electroluminescence imaging calibrated against local series-resistance maps during coating would isolate interconnect losses. Cross-study evaluation of interfacial stabilization failed because studies reported initial unmasked voltages rather than quantifying interfacial bonding enthalpies alongside continuous operational fill-factor retention under bias.

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Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability Statement

The corpus table for the month, the per-paper extraction records with their verbatim source quotations, the statistics file underlying every number in the text, and the gate report are provided as Supplementary Information. Abstract text and full texts are not redistributed.

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